are the source. On the other hand, point of utilization is called sink. According to Munch mass flow hypothesis, the solutes move along the concentration gradient in a bulk flow.

Although minerals are dissolved in soil water, they do not tend together with water to enter the root hairs during absorption of water. Mineral absorption is independent of water absorption. Minerals are absorbed either actively or passively.

Evaluation

1. In a fully turgid cell
   a. DPD = 10 atm; OP = 5 atm; TP = 10 atm
   b. DPD = 0 atm; OP = 10 atm; TP = 10 atm
   c. DPD = 0 atm; OP = 5 atm; TP = 10 atm
   d. DPD = 20 atm; OP = 20 atm; TP = 10 atm

2. Which among the following is correct?
   i. apoplast is fastest and operate in nonliving part
   ii. Transmembrane route includes vacuole
   iii. symplast interconnect the nearby cell through plasmadesmata
   iv. symplast and transmembrane route are in living part of the cell

   a. i and ii b. ii and iii
   c. iii and iv d. i, ii, iii, iv

3. What type of transpiration is possible in the xerophyte *Opuntia*?
   a. Stomatal b. Lenticular
   c. Cuticular d. All the above

4. Stomata of a plant open due to
   a. Influx of K^+ b. Efflux of K^+
   c. Influx of Cl^- d. Influx of OH^-

5. Munch hypothesis is based on
   a. Translocation of food due to TP gradient and imbibition force
   b. Translocation of food due to TP
   c. Translocation of food due to imbibition force
   d. None of the above

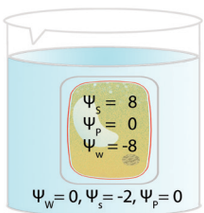
6. If the concentration of salt in the soil is too high and the plants may wilt even if the field is thoroughly irrigated. Explain

7. How phosphorylase enzyme open the stomata in starch sugar interconversion theory?

8. List out the non-photosynthetic parts of a plant that need a supply of sucrose?

9. What are the parameters which control water potential?

10. An artificial cell made of selectively permeable membrane immersed in a beaker (in the figure). Read the values and answer the following questions?



EAPK4

   a. Draw an arrow to indicate the direction of water movement
   b. Is the solution outside the cell isotonic, hypotonic or hypertonic?
   c. Is the cell isotonic, hypotonic or hypertonic?
   d. Will the cell become more flaccid, more turgid or stay in original size?
   e. With reference to artificial cell state, is the process endosmosis or exosmosis? Give reasons

XIth_Botany_Chapter_11_EM.indd 257 05-01-2023 18:29:12